

Failure 1 — the recall ceiling

The reranker can't fix what it never sees

QUERY

*"romantic agriturismo
near Florence with a vineyard"*

THE PERFECT LISTING

exists in inventory · ranked #1,400 by first-stage

Casa di Lucia

listing description (truncated)

"family-run estate in the
Chianti hills, working winery,
two-bedroom suite, terrace
overlooking the vines..."

- never says "romantic"
- never says "agriturismo"

vocabulary mismatch · BM25 misses · dense ranks #1,400

FIRST-STAGE RANK

BM25 / dense over corpus

rank	candidate
#1	
#2	
#3	
...	
#100	
...	
#1000	
reranker cutoff	
#1001	
...	
#1400	Casa di Lucia
...	
#5000	
...	

Ncandidates · sorted by lexical / dense score

top-1000 sent to rerank

RERANKER

*cross-encoder /
reasoning model*

budget: top-1000 only

*excellent — but blind to
what's not in the input*

never reaches reranker

A perfect document, lost forever between the retriever and the reranker.

A better reranker doesn't help. The relevant doc was never on the table.